Lecture 5 :

Recap last week we talked about divisibility

$$\rightsquigarrow \quad n = \prod_{\substack{p \text{ prime} \\ p \mid n}} p^{PExp(n)(p)}.$$

Today we'll talk about calculating modulo a numbers.

Eucledean division: n, m two integers, $n \neq 0$. ↗ nat. numbers.

Then there exist unique integers k, r s.t.

$$m = k \cdot n + r \qquad (0 \leq r < n)$$

Defn In the case above we say $m \equiv r \ (\mathrm{mod}\ (n))$.

Observe: If $m_1 \equiv r_1 \ \mathrm{mod}\ (n)$ then $m_1 + m_2 \equiv r_1 + r_2 \ \mathrm{mod}\ n$.

$m_2 \equiv r_2 \ \mathrm{mod}\ (n)$

Objection: $r_1 + r_2$ might be larger than n.

So the set $\{0, \dots, n-1\}$ is not a good place/setting to talk about calculations mod n!

Defn $\mathbb{Z}/n = \{$ set of eq. classes of $m \in \mathbb{Z}$ w.r.t. to the equivalence relation $m \sim m' \Leftrightarrow n \mid m - m'$ $\}$.

Quick reminder / introduction on equivalence relations.

Def. Let X be set (type). Then an *equivalence relation* is function $R : X \times X \longrightarrow \mathrm{Prop}$

$$(x,y) \longmapsto x \sim_R y \,.$$

s.t.
- $\forall x \in X : \quad x \sim_R x$. (reflexive)
- $\forall x, \forall y \in X : \quad x \sim_R y \iff y \sim_R x$ (symmetric).
- $\forall x, y, z \in X : \quad x \sim_R y \wedge y \sim_R z \implies x \sim_R z$. (transitivity)

Ex. $R(x,y) = (x = y)$

Def. Let R be an eq. relation on X and $x \in X$. Then the equivalence class of x, is

$$[x]_R = \{ y \in X \mid x \sim_R y \}.$$

Exc In the sit. above. We have

$$[x]_R = [y]_R \iff x \sim_R y \,.$$

Back to the integers

We define the following relation on $\mathbb{Z}$.

$$R_n : \mathbb{Z} \times \mathbb{Z} \longrightarrow \mathrm{Prop}$$

$(m_1, m_2) \longmapsto n \mid m_1 - m_2$.

<u>Defn</u> $\mathbb{Z}/n = \{ [m]_{R_n} \mid m \in \mathbb{Z} \}$. $[m]_n$

<u>Ex</u>. • The eq. class of $[0]_n = \{ k \cdot n \mid k \in \mathbb{Z} \}$.

<u>Why</u>. $m \in [0]_n \leftrightarrow n \mid m - 0 \Leftrightarrow n \mid m$.

<u>Did we solve our problem?</u> Yes

$m \underset{n}{\sim} r \qquad m = nk + r$

1) If $m \equiv r \pmod{n}$ then $[m]_n = [r]_n$.

2) $m \in \mathbb{Z}$, $m' \in \mathbb{Z}$. $k \in \mathbb{Z}$.

$$[m+k]_n = [m'+k]_n \quad \text{if} \quad [m]_n = [m']_n.$$

Combining these let's us define an addition for equivalence <u>classes</u>.

$$[m]_n + [k]_n := [m+k]_n.$$

<u>Upshot</u>: If $m_1 \equiv r_1 \pmod{n}$
$m_2 \equiv r_2 \pmod{n}$
Then <u>$[m_1+m_2]_n = [r_1]_n + [r_2]_n$</u>.

Observation: If $n_1 \mid n_2$ then we have

$$m_1 \sim_{n_2} m_2 \implies m_1 \sim_{n_1} m_2 .$$

$$\iff n_2 \mid m_1 - m_2 \implies n_1 \mid m_1 - m_2 .$$

This allows us to to define a map.

$$\begin{aligned} \mathbb{Z}/_{n_2} &\longrightarrow \mathbb{Z}/_{n_1} \\ [x]_{n_2} &\longmapsto [x]_{n_1} . \end{aligned}$$

(If $[x]_{n_2} = [y]_{n_2}$ then $[x]_{n_1} = [y]_{n_1}$)

Fix $n, m \in \mathbb{Z}$. $(n > 0)$
In part, we have two maps

$$\begin{tikzcd} \mathbb{Z}/_{m*n} \arrow[r] \arrow[d] & \mathbb{Z}/_m \\ \mathbb{Z}/_n & \end{tikzcd} \qquad \begin{tikzcd} {[x]_{m*n}} \arrow[r, mapsto] \arrow[d, mapsto] & {[x]_m} . \\ {[x]_n} & \end{tikzcd}$$

Combine the two maps: $C_{mn} : \mathbb{Z}/_{m*n} \longrightarrow \mathbb{Z}/_m \times \mathbb{Z}/_n$

$$[x]_{m*n} \longmapsto ([x]_m, [x]_n).$$

Theorem (Chinese remainder theorem) Let $m, n \in \mathbb{Z}$.
Assume $n > 0$, $m > 0$ and $\gcd(n, m) = 1$.

Then $C_{mn} : \mathbb{Z}/_{m*n} \longrightarrow \mathbb{Z}/_m \times \mathbb{Z}/_n$ is a
$[x]_{m*n} \longmapsto ([x]_m, [x]_n)$
bijection.

Quick recap on functions.

Let $f : X \to Y$ be a function then we say

- f is injective if $\forall x, y \in X : f(x) = f(y) \Rightarrow x = y$.
- f is surjective if $\forall y \in Y\ \exists x \in X$ s.t. $f(x) = y$.
- f is bijective if f injective $\land$ f surjective.
  (this mean 1-1 mapping)

Lemma: $f : X \to Y$, and X and Y are finite sets.
Then (f injective $\land$ $|X| = |Y|$) $\Rightarrow$ f surjective

Proof. Pigeon hole principle?

Proof. By contradiction assume f is surjective.

Then $\exists y \in Y$ s.t. $\forall x \in X$ $f(x) \neq y$.

In other words image of f $\subset Y \setminus \{y\}$.

$|$ — $| \leq |X|$ $| < |Y|$

$(= |Y|$.

Proof of the theorem

We want to apply the lemma.

Step 0) Show that $\mathbb{Z}/n$ is finite!

Idea: Consider the map

$$q : \{0, \dots, n-1\} \xrightarrow{\sim} \mathbb{Z}/n$$
$$m \longmapsto [m]_n.$$

Claim. q is surjective!

Why: For every $m \in \mathbb{Z}$, we have $[m]_n = [r]_n$ where r was the remainder after eucl. division.

Step 1: q is also injective!

Claim. if $m \neq m' \in \{0, \dots, n-1\}$ then $[m]_n \neq [m']_n$

$\Leftrightarrow m \sim_n m'$.

$|m - m'| < n$.

Because

Thus, $n \mid m-m' \Rightarrow m-m' = 0$.

<u>Consequence</u>: $|\mathbb{Z}/_n| = n$.

For the theorem: $m * n = |\mathbb{Z}/_{m*n}|$

$$m * n = |\mathbb{Z}/_m| * |\mathbb{Z}/_n|$$
$$= |\mathbb{Z}/_m \times \mathbb{Z}/_n|.$$

<u>Step 2</u>: Show that C_{mn} is injective!

Let $[x_1]_{nm}, [x_2]_{nm} \in \mathbb{Z}/_{m*n}$.

<u>Assume</u>
$$\begin{array}{ccc} C_{mn}([x_1]_{nm}) & = & C_{mn}([x_2]_{nm}) \\ \| & & \| \\ ([x_1]_n, [x_1]_m) & = & ([x_2]_n, [x_2]_m) \end{array}$$

<u>Want to show:</u> $[x_1]_{nm} = [x_2]_{nm}$. ($\Leftrightarrow x_1 \sim_{m*n} x_2$).

From the assumption we get:
$$\underline{n \mid x_1 - x_2} \quad \text{and} \quad m \mid x_1 - x_2.$$

The goal translates to $m*n \mid x_1 - x_2$.

In fact, this follows from the following lemma

<u>Lemma</u> Assume $\gcd(m,n)=1$ and $m \mid n*k$

then $m \mid k$.

<u>Pf.</u> Assume $m \nmid k$. then $\exists$ prime number p

s.t. $p \mid m$ but $p \nmid k$.

then $p \mid n*k \Rightarrow p \mid n$.

$\Rightarrow p \mid \gcd(m,n)$ $\lightning$.